

A new Web Platform API

WebRTC Encoded Source

Leonardo Evi

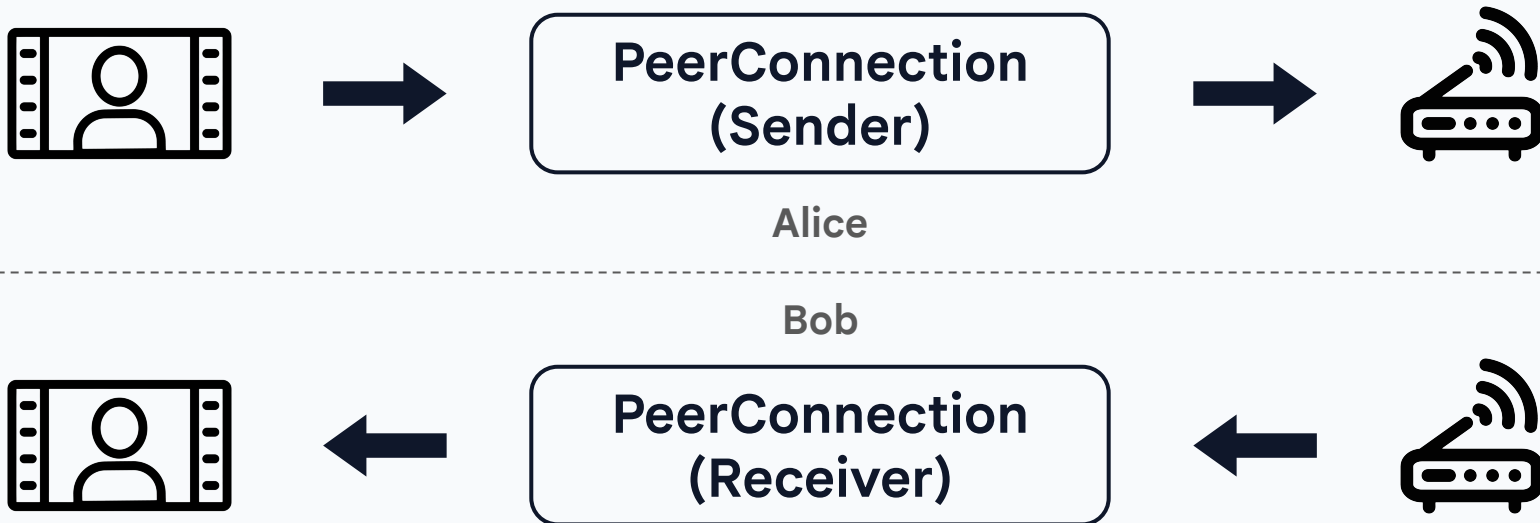
evil@chromium.org | leonardoevi@google.com





The PeerConnection

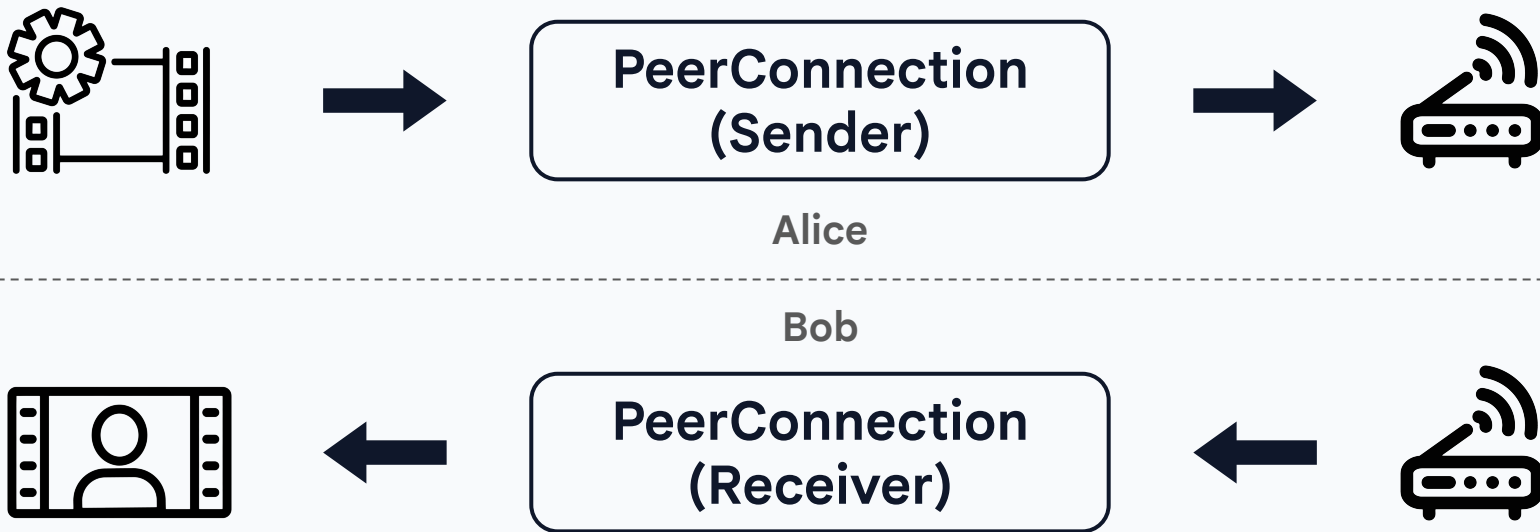
- The Web Platform exposes the PeerConnection (PC) API
- PC is designed to accept raw media via a track, encode it, and send it





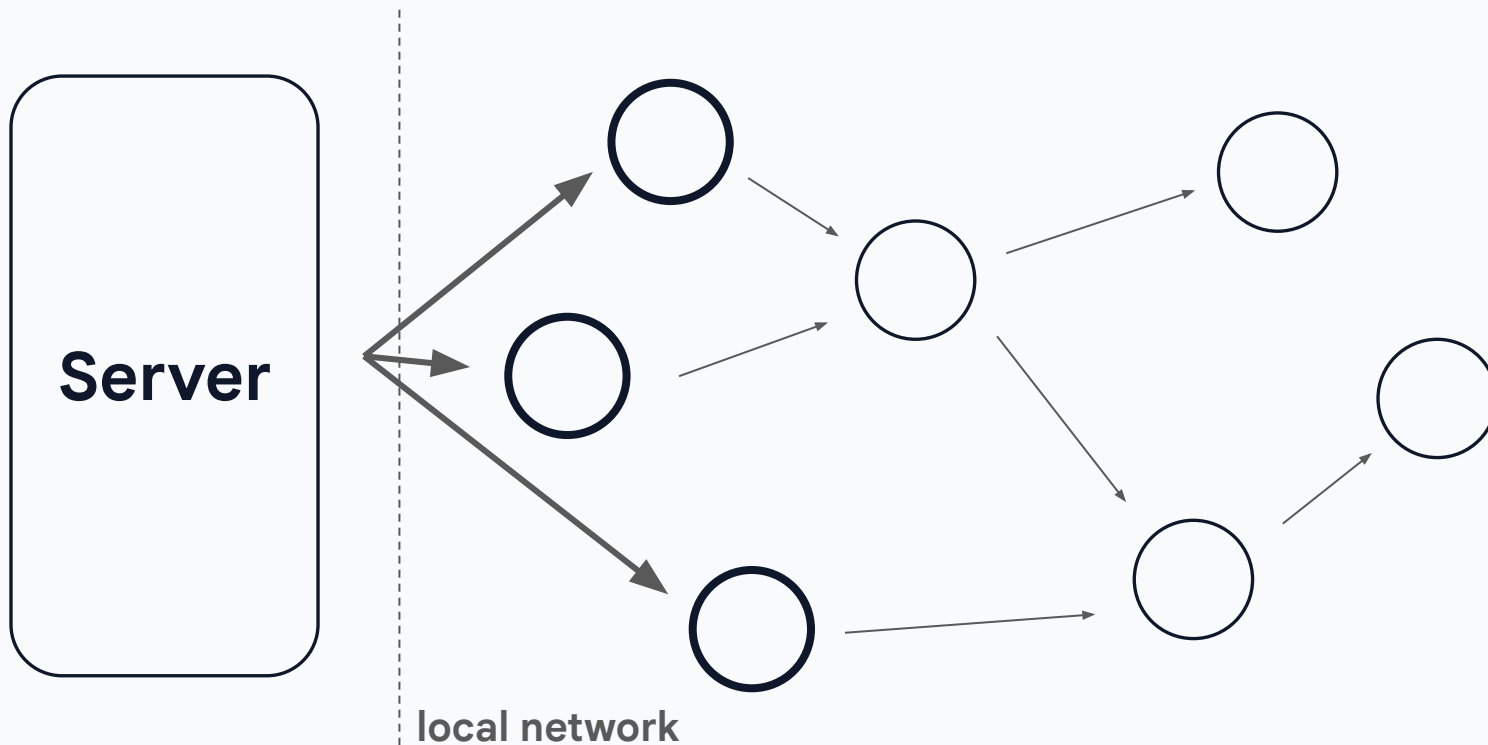
Injecting Encoded Media

- Some Applications require the PeerConnection to send already-encoded media





Peer Assisted Media Forwarding

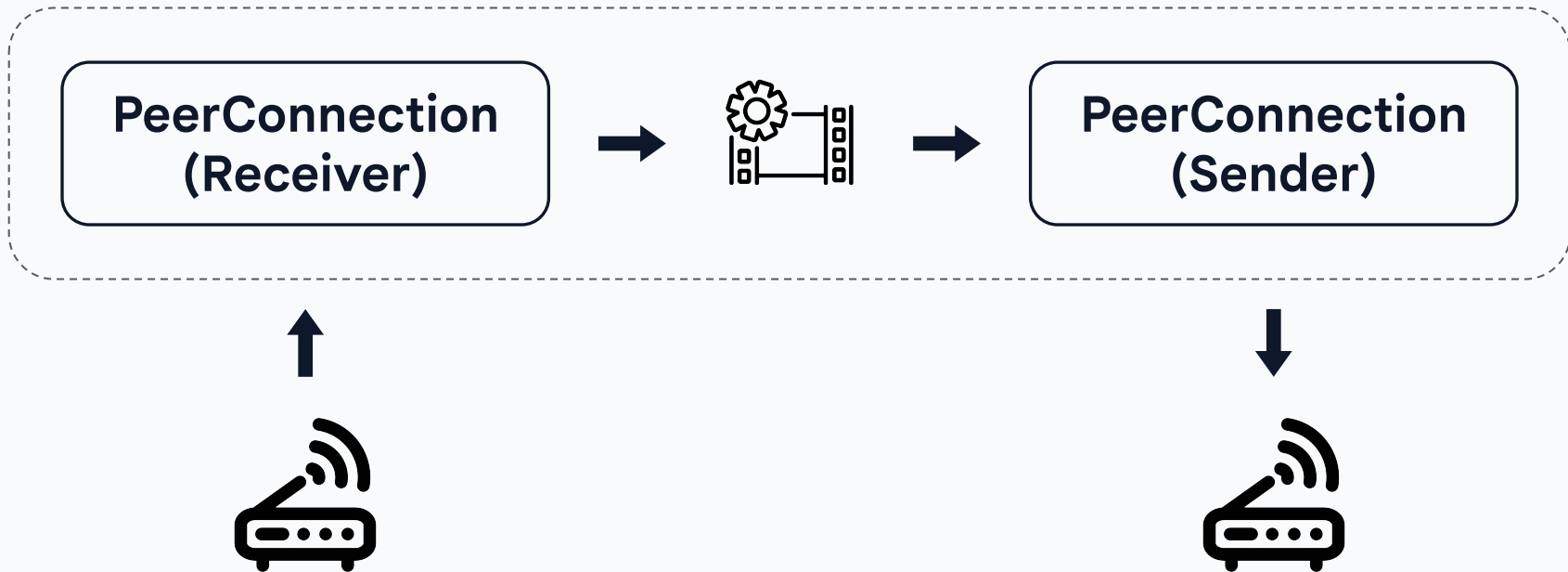




Peer Assisted Media Forwarding

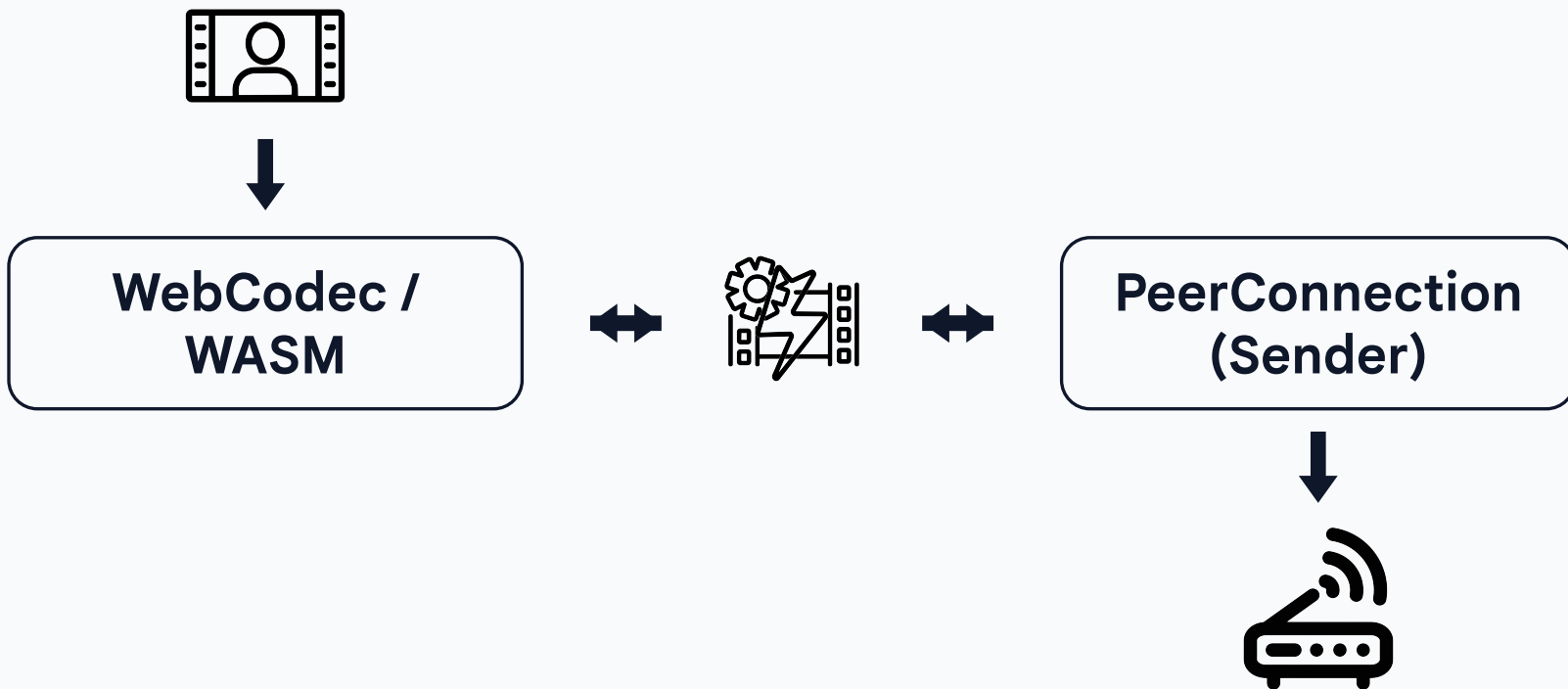
- Avoids re-encoding

N-th Node





Custom Codec + PeerConnection





Injecting Encoded Media

The Requirement

Encoded Frame Injection

- An API built explicitly to act as a **source of encoded frames** for a PeerConnection.
- Replaces Media Track
- **Bi-directional** WebRTC signaling support for bitrate limits and keyframe requests.

Existing APIs

Encoded Transform API

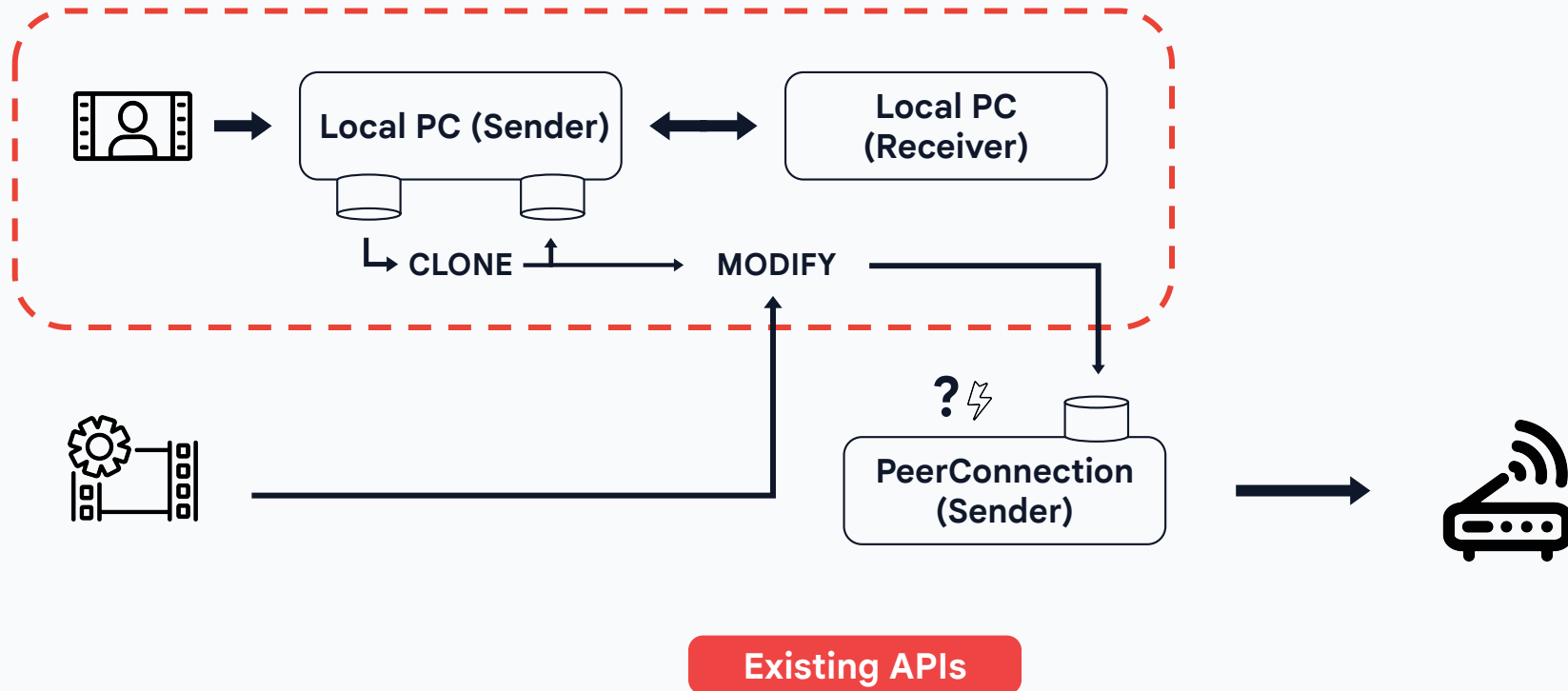
- Designed exclusively for **Read-Modify-Write** pipelines (e.g., end-to-end encryption).

Legacy Insertable Streams API

- Chrome only, Non Standard
- Unergonomic
- Limited functionality



The setup for Injection





Create Custom Encoded Video Frames

```
const frame = new RTCEncodedVideoFrame({  
  type: 'key',  
  payloadType: 96,  
  mimeType: 'video/VP8',  
  rtpTimestampWithoutOffset: 54321,  
  width: 640,  
  height: 480,  
  captureTime: performance.now(),  
  contributingSources: [1234],  
  data: buffer,  
})
```



Create Custom Encoded Audio Frames

```
const frame = new RTCEncodedAudioFrame({
  contentType: 'speech',
  payloadType: 111,
  mimeType: 'audio/opus',
  rtpTimestampWithoutOffset: 54321,
  captureTime: performance.now(),
  contributingSources: [1234],
  audioLevel: 0.69,
  data: buffer,
})
```



Encoded Source Object

```
// On main window
await sender.createEncodedSource(worker)

// On web worker
self.onrtcsenderencodesource = (event) => {
  encodedSource = event.encodedSource
  writer = encodedSource.writable.getWriter()

  while(...)
    writer.write(frame)
}
```

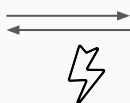
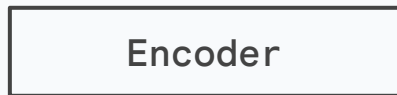


Encoded Source Object

```
// On worker
self.onrtcsenderencodesource = (event) => {
  [...]
  encodedSource.onkeyframerequest = (e) => {
    writer.write(keyFrame)
  }
  encodedSource.onbitrateinfochange = (e) => {
    log(encodedSource.allocatedBitrate)
    log(encodedSource.availableOutgoingBitrate)
  }
}
```



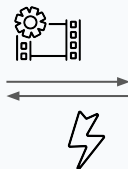
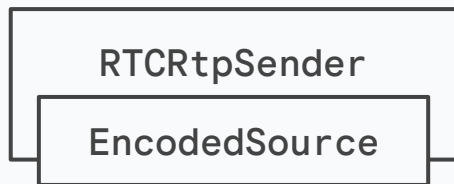
Standard Media Processing



chromium
webrtc



Encoded Source



chromium
webrtc

Solution

WebRTC Encoded Source API



Live



Demo